Gautham Ramkumar

SLAM, Localization & State Estimation Engineer

Boston, MA • ramkumar.g@northeastern.edu • gautham-ramkumar.github.io • linkedin.com/in/rgautham20 • github.com/gautham-ramkumar

F-1 STEM OPT eligible; no immediate sponsorship required

Summary

Robotics M.S. candidate focused on localization for GPS-denied autonomy, working on the sensor fusion, calibration, and factor-graph layers that keep a robot's pose estimate reliable under noise, latency, and signal loss.

Education

Northeastern University, Boston, MA **2024 – 2026**

M.S. Robotics — Concentration in Electrical & Computer Engineering

- **Coursework:** Advanced Perception, Robot Sensing & Navigation, Autonomous Field Robotics, Mobile Robotics, Robot Mechanics & Controls, Legged Robotics

Vellore Institute of Technology (VIT), Chennai, India **2020 – 2024**

B.Tech. Electronics & Communications Engineering

Technical Skills

Languages: Python, C++, MATLAB
State Estimation & Calibration: Extended Kalman Filters, Factor Graph Optimization, Bundle Adjustment, Multi-Sensor Calibration, Camera-LiDAR Calibration, Temporal Synchronization, Sensor Fusion, Pose Estimation
Perception & Vision: Visual SLAM (RTAB-Map), Stereo Vision, Depth Estimation, Object Detection (YOLO), 3D Reconstruction (SfM), Feature Matching (SIFT/ORB), Point Cloud Processing, Loop Closure Detection
Frameworks & Tools: ROS2, PyTorch, OpenCV, GTSAM, Open3D, TensorRT, CUDA, COLMAP, Docker, Git
Hardware & Sensors: ZED Mini Stereo Camera, Intel RealSense D435, LiDAR, VectorNav IMU, NVIDIA Jetson Nano

Selected Projects

Camera-LiDAR Temporal Calibration — -19.49 ms offset, stable to 1.5 ms **Oct 2025 – Dec 2025**

Python · C++ · GTSAM · ROS2 · Open3D · OpenCV

- Built a stamp-primary temporal synchronization pipeline anchored on hardware capture timestamps, with a bounded residual path correcting latency bias via cross-modal edge alignment.
- Estimated a stable -19.49 ms camera-LiDAR offset across 1,500 camera frames and 750 LiDAR scans, repeatable to within 1.5 ms across windows.

Indoor 3D Mapping with RTAB-Map — 90% ground-truth alignment, GPS-denied **Sep 2025 – Oct 2025**

ROS2 · ZED Mini · LiDAR · VectorNav IMU

- Deployed RTAB-Map visual SLAM on a ZED Mini stereo camera in ROS2, bringing up a full sensor suite (LiDAR, VectorNav IMU) with calibrated coordinate transforms.
- Validated loop-closure detection and pose-graph consistency across long trajectories in repetitive, GPS-denied tunnels.

GPS-IMU Fusion via Extended Kalman Filter — 600 Hz IMU prediction rate **Sep 2025 – Oct 2025**

Python · NumPy

- Implemented a real-time 2D vehicle localization EKF fusing a 600 Hz IMU with asynchronous, low-rate GNSS measurements.
- Propagated state through IMU kinematics and weighted GPS position updates by measurement uncertainty.

3D Reconstruction via Structure from Motion — ~60-70% BA cost reduction **2025**

Python · OpenCV · GTSAM · Open3D

- 'Built an incremental SfM pipeline from scratch': 'SIFT features, RANSAC outlier rejection, essential-matrix decomposition, and triangulation.'
- Optimized roughly 5,500 landmarks through GTSAM factor-graph bundle adjustment, cutting bundle adjustment cost 60-70%.

Experience

SmartInternz **2023**

IoT Engineering Intern

- Built real-time sensing pipelines and cloud logging systems on IoT hardware.
- Integrated embedded sensor data acquisition with cloud infrastructure.

Awards

2nd Place, Revolute Hackathon (Aug 2026) — Voice-controlled robotic manipulation for cooking, built in a weekend.